

# AAYUSH KUMAR

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## Education

### The National Institute of Engineering

Bachelor of Engineering in Computer Science and Engineering (AI & ML)

Aug 2024 – May 2028

CGPA: 9.3/10

## Technical Skills

**Languages:** C++, JavaScript (ES6+), TypeScript, Python, SQL, Java  
**Frontend:** React.js, Vite, Tailwind CSS, HTML5, CSS3, Framer Motion  
**Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, WebSockets, Server-Sent Events (SSE), BullMQ  
**Databases:** PostgreSQL, MongoDB, Redis, Firebase  
**DevOps & Tools:** Docker, Linux (Fedora), Git, GitHub, GitHub Actions, CI/CD, Postman, Vercel, Render  
**Core CS:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

## Projects

### Auralis

Jul 2026

*React.js, TypeScript, FastAPI, LangGraph, PostgreSQL, Redis, FAISS, Docker*

[GitHub](#) | [Live](#)

- Built an **AI-powered sales intelligence platform** using LangGraph that classifies objections with **94% confidence**, detects personas and sentiment, and generates contextual responses in under 2 seconds.
- Implemented a **RAG pipeline** with FAISS and a production-ready stack using React, FastAPI, PostgreSQL, Redis, JWT, WebSockets, Docker, and GitHub Actions CI/CD.
- Designed a modular **multi-agent architecture** with strategy-specific routing, Redis-backed session management, analytics tracking, and scalable deployment on Vercel and Render.

### Beatzy

Apr 2026

*Python, FastAPI, React.js, Node.js, PostgreSQL, Redis, BullMQ, HuggingFace*

[GitHub](#) | [Live](#)

- Developed a full-stack **music intelligence platform** supporting BPM, key, chord, mood analysis, lyric synchronization, and metadata enrichment using librosa and YAMNet.
- Designed an asynchronous pipeline with **BullMQ, Redis, and PostgreSQL**; deployed microservices on Render and Vercel.
- Integrated external music metadata services and optimized asynchronous processing with Redis caching, improving response latency for repeated analyses.

### DeskGuard

May 2026

*React.js, Node.js, Express.js, PostgreSQL, Redis, SSE, Docker*

[GitHub](#) | [Live](#)

- Built a **library seat booking platform** with live desk maps, QR check-ins, and automated seat reclamation workflows.
- Implemented **Server-Sent Events (SSE)** for real-time desk availability without polling, ensuring synchronized updates across connected clients.
- Designed PostgreSQL audit logging and Redis TTL-based session management for automatic seat recovery and reliable booking workflows.

### Citizen Resolver System

Mar 2026

*React.js, Vite, Tailwind CSS, REST APIs, JWT*

[GitHub](#) | [Live](#)

- Built a **civic grievance platform** for complaint reporting, tracking, and department-level workflow management.
- Implemented JWT-authenticated CRUD, reusable React components, and responsive dashboards with real-time status updates.
- Structured REST API endpoints and role-based access so citizens and department staff see distinct, permission-scoped views of the same complaint data.

## Experience

### OWASP NIE Student Chapter

Aug 2025 – Present

*President*

*NIE, Mysore*

- Lead the chapter's **technical initiatives, developer events, and security-focused programs**, coordinating student teams and supporting 100+ participants across campus events.
- Organize and deliver technical workshops on **Git, GitHub, open source, web development, and cybersecurity**, helping students build practical development skills.
- Oversee technical infrastructure, event workflows, and collaborative development processes while mentoring members across the chapter's technical teams.

### Open Source Contributor

2025 – Present

*Independent*

*GitHub*

- Build and maintain **production-grade full-stack and ML applications** spanning React, Node.js, FastAPI, PostgreSQL, Redis, and modern AI/ML pipelines.
- Maintain deployment-ready repositories with **Docker, GitHub Actions, and CI/CD workflows**, emphasizing reproducibility, documentation, and scalable architecture.

## Achievements

- **Hackathons:** **3× Hackathon Finalist**; secured **3rd Place** at ThinkRoot x Vortex'26 (NIT Trichy), competing against 100+ teams and building [LeadForge](#) in 24 hours.
- **Competitive Programming:** Solved **600+ DSA problems** across platforms, covering Graphs, Dynamic Programming, Trees, and Advanced Data Structures.
- **Open Source:** Maintain a portfolio of **full-stack and ML applications** with production-oriented architecture, Docker-based environments, and CI/CD workflows.